Moligo Technologies – Investment Memo

Moligo is a bet on a newer DNA-making technology replacing the current standard method, specifically for producing long strands of DNA, where the current approach runs into hard limits. The main question is whether Moligo can manufacture at the scale and quality that drug companies need.

Core investment question: Can Moligo's enzyme-based DNA production match the cost and quality of today's standard method at manufacturing scale within 5–7 years?

1. Company Snapshot

Moligo is a privately held Swedish biotech (~16 employees, founded 2019) developing an enzymatic platform for long single-stranded DNA (ssDNA) synthesis for gene editing and cell therapy applications. The company is pre-Series A with limited disclosed funding (<$1M estimated).

Funding estimates range from $170K–$782K (Crunchbase / PitchBook). Investors include Colibri Ventures, Vento, Nordic Science Capital, EIT Health, and Phase2Phase Biopharma.

Key observation: Moligo is significantly less capitalized than leading competitors despite early scientific validation.

2. Core Problem

DNA synthesis constraints

Today's standard method for making synthetic DNA works well for short pieces, but starts breaking down as you try to make longer ones:

- Can only reliably make DNA up to about 200–300 building blocks (base pairs) long
- Longer sequences get increasingly expensive and error-prone
- Accuracy drops further when the DNA sequence is complex or has repeating patterns

Why it matters

Gene editing, cell therapy, and other cutting-edge treatments increasingly need long, accurate DNA pieces that the current method can't reliably produce.

Economic constraint

With today's method, cost rises sharply as DNA gets longer, because each added building block requires a separate chemical step and errors stack up. An enzyme-based approach could keep costs flat regardless of length but only if accuracy and output hold up at larger scales.

Quantified Technical Constraint + Unit Economics Baseline

2.1 How the two methods compare

Parameter	**Phosphoramidite (baseline)**	**Enzymatic (target)**	**Gap**
Max practical length	~200–300 bp	>10,000 bp	+30–50x
Cost per kb (GMP)	~$1,000–50,000+	unknown / target range	?
Step yield	high but cumulative loss	unknown	?
Error profile	substitution + truncation	enzymatic fidelity unknown	?
GMP maturity	mature	pre-GMP	large gap

The core advantage Moligo is chasing: today's method gets more expensive with every DNA building block added, because each one requires its own chemical reaction. Moligo's approach has a higher upfront cost, but then copies DNA from a template similar to how cells do it naturally. If Moligo can do this while keeping errors low and meeting pharmaceutical quality standards, those are the two things they need to prove to have a real business.

3. Technology & Scientific Differentiation

Moligo's platform can produce long single-stranded DNA (over 10,000 building blocks) without using the copying step (PCR) that most methods rely on. It can make both straight and circular DNA.

The technology was tested in a study published in Nature Communications, where it was used with Cellectis (a gene-editing company) and showed better results for inserting genes than existing approaches.

Why it could work

- Can deliver large gene sequences that current methods can't handle
- Less toxic to cells than virus-based gene delivery methods
- Fits into newer gene editing approaches that avoid using viruses

3.1 How the production process works

Build template → enzymes copy and extend it → control which sequence gets made → stabilize the DNA strand → purify it → verify quality → scale up to pharmaceutical manufacturing standards.

Key risks

- No one has yet proven it works at the scale and quality standards pharmaceutical companies require
- The whole process depends on first making a highly accurate master template which itself is difficult

- Very long single-stranded DNA tends to fold in on itself, and the enzyme doing the copying may stall before finishing
- Some DNA strands don't get fully made, and the purification process can further reduce how much usable product you end up with

4. Market & Competitive Landscape

Industry structure

The market for long DNA and non-viral gene editing is shaping up as one where winning comes down to who can manufacture reliably at scale and get regulatory approval — not just who has the cleverest science.

The field is moving toward:

- Delivering genes without using viruses, in cases where that's now possible
- Greater use of long DNA pieces as the "repair instructions" in CRISPR gene editing
- A gradual shift from purely chemical DNA production to enzyme-based and mixed approaches

Competitive axes

Who wins will come down to:

- Meeting pharmaceutical manufacturing quality standards (GMP)
- Ability to produce large quantities at a reasonable and stable cost
- How long a DNA piece they can reliably make
- Progress getting regulatory sign-off
- Partnerships or supply deals with pharmaceutical companies

Across all of these, making it consistently and at scale matters more than having a slightly different enzyme design.

Competitor positioning

Company	Positioning	Advantage	Risk
Touchlight	Circular ssDNA manufacturing platform	GMP maturity, pharma integration	Regulatory expansion still evolving
Ansa Biotechnologies	Enzymatic DNA synthesis	Strong technical leadership	Commercial scaling uncertainty
Kano Therapeutics	Therapeutic circular ssDNA	Direct clinical integration	Early execution risk
Full Circles Therapeutics	Circular ssDNA engineering	Strong early validation	Clinical translation risk

Company	Positioning	Advantage	Risk
GXstrands / CPTx	Fermentation-based ssDNA	Potential cost advantage	Platform immaturity

Moligo positioning

Strengths

- Strong academic and scientific credibility
- Demonstrated long ssDNA capability
- Early gene editing validation
- European academic-industry ecosystem

Weaknesses

- Limited capitalization vs US peers
- No established GMP manufacturing
- Early commercial infrastructure
- Scale-up not yet de-risked

Structural insight:
The market will probably settle on a small number of platforms that can manufacture at pharma grade, get regulatory acceptance, and land supply deals. Whoever executes first at scale wins.

4.1 How the market could look in 5–7 years

Scenario A: Oligopoly (Most Likely)

- The market narrows to 2–4 dominant DNA manufacturing platforms, likely including established players like Touchlight plus one or two successful enzyme-based newcomers.
- Regulatory approval history, manufacturing quality track record, and clinical experience create high barriers that make it hard for new entrants to compete.
- Once a DNA supplier is built into a drug company's development and manufacturing process, switching is expensive and slow, creating loyalty by default.
- Providers compete on reliability, scale, and supply security rather than just price.
- This would look similar to the contract drug manufacturing industry, where a handful of trusted suppliers take most of the value.

Scenario B: Fragmented Tooling Market

- Many providers each offer a different DNA production approach for different niches.
- Technologies get picked up mainly in research and early-stage work, without any single one becoming the go-to standard for pharmaceutical manufacturing.

- Customers choose based on their specific needs, length, speed, cost, or application.
- No single approach gets far enough ahead to dominate.
- Suppliers make money through specialist services and tools, not large-scale drug manufacturing.

Scenario C: Vertical Integration

- Large pharma and gene-editing companies bring DNA production in-house once the technology is mature enough.
- DNA manufacturing becomes a core internal capability, like how some drug companies handle their own biologics production.
- External suppliers shift to selling the tools, enzymes, and equipment rather than finished DNA.
- The edge moves to process expertise, analytics, and know-how rather than just producing DNA.
- Suppliers become tool providers while drug companies own their production.

Implications for Moligo

- How much Moligo is worth long-term depends heavily on whether the market consolidates (Scenario A) or stays fragmented (Scenario B).
- In Scenario A, Moligo could build a strong, lasting position if it validates its platform early enough.
- In Scenario B, the technology still has value but profits would be harder to protect as many competitors coexist without a clear winner.

5. Moat Analysis

Moligo's competitive protection can be thought of in three layers:

1. Patents (weakest layer)

It's not clear how strong or broad Moligo's patents are. This probably isn't where the real protection comes from.

2. Manufacturing know-how (core layer)

The real edge likely comes from:

- fine-tuning how the enzymes work
- proprietary purification steps
- getting consistent results batch after batch
- keeping accuracy high over very long DNA sequences

3. Manufacturing scale (strongest layer)

If Moligo gets there first, the combination of certified manufacturing, optimized yields, and regulatory approval becomes very hard for a late entrant to replicate quickly..

6. How the thesis could break down

Layer	Primary Failure Risk
Molecular	The DNA comes out with errors or incomplete sequences at the lengths needed for actual therapies
Manufacturing	The cost of cleaning up and recovering the DNA erases the price advantage over the current method
Commercial	Drug companies don’t switch even if the technology works, because the switching cost and risk aren’t worth it to them

7. Key Catalysts

Near-term

- Series A financing
- Strategic pharma partnerships
- Additional validation studies
- Progress toward GMP manufacturing

Medium-term

- Clinical adoption of ssDNA donor templates
- Expansion into pharmaceutical manufacturing supply chains
- Regulatory milestones in non-viral gene editing

8. Investment Thesis

Moligo is a high-risk, high-reward bet on becoming core infrastructure for next-generation DNA manufacturing.

The central bet is that Moligo’s enzyme-based production can move from a lab curiosity to an industrial-scale, pharmaceutical-grade process, and in doing so, replace the current standard method for making long DNA.

The main risk isn’t whether the science works, early results suggest it does. It’s whether the company can make it work at scale, consistently, and at a cost that makes commercial sense.

Rough estimate: a 20–30% chance of successfully scaling to pharmaceutical-grade manufacturing, but with potential returns of 10x or more if it gets there.

9. When to change your mind

The investment case weakens materially if several of these are still true after 24 months:

- Moligo still hasn't shown it can reliably make DNA over 10,000 building blocks long at the accuracy levels gene editing requires
- No sign that costs are tracking toward being competitive with today's standard method for longer DNA
- No realistic path to pharmaceutical-grade manufacturing, and no manufacturing partner, within 18–24 months
- No partnership or supply deal with a drug company in gene editing
- A competitor like Touchlight or another enzyme-based player pulls clearly ahead on cost and output at the same scale

If two or more of these are still true after 24 months, the investment case is significantly weaker.

10. Bull Case

- Proves scalable DNA production at the lengths and quality levels clinical applications need
- Becomes the go-to DNA supplier for non-viral gene editing programs
- Expands into synthetic biology and cell therapy manufacturing beyond gene editing
- Becomes a foundational supplier in genetic medicine, the provider of the backbone onto which CRISPR-Cas9 and other technologies rely on

11. Bear Case

- Can't achieve stable, pharmaceutical-grade production at acceptable quality and output
- Gets outpaced by better-funded competitors like Touchlight or other enzyme-based players
- The industry sticks with existing DNA delivery methods (viruses, plasmid DNA) and adoption of Moligo's approach is slow
- Non-viral gene editing doesn't take off as fast as expected
- Runs out of money before proving the technology at industrial scale

12. Conclusion

The central question is whether Moligo can prove pharmaceutical-grade, scalable DNA production before competitors lock up the manufacturing relationships and regulatory approvals.

In the near term, the case gets meaningfully stronger with at least one of:

- A certified pharmaceutical-grade manufacturing facility or partner
- A signed supply or co-development deal with a drug company
- A second independent study confirming the long DNA performance in a different therapeutic application

Without progress on at least one of these within 18–24 months, the risk of being overtaken by better-funded competitors becomes the dominant concern.